

Homework 4

1.7.4 Let $a > 1$ and $a \notin \mathbb{Z}$. Prove that $|x|^a$ lies in $\mathcal{C}^{[a]}(\mathbb{R}^n) \setminus \mathcal{C}^{[a]+1}(\mathbb{R}^n)$.

Proof. We will start by proving a critical lemma regarding the form of partials of $|x|^a$.

Lemma 1. All partials of $|x|^a$ can be expressed as $\partial^\alpha |x|^a = \sum_{i \in I} C_i x^{\alpha_i} |x|^{P_i}$, where α, α_i are multi-indices. Furthermore, define $\text{ord}(C_i x^{\alpha_i} |x|^{P_i}) = |\alpha_i| + P_i$. Then each partial taken will reduce $\text{ord}(C_i x^{\alpha_i} |x|^{P_i})$ by exactly 1.

Proof. We proceed by induction. For the base case, take a partial of $|x|^a$ in some variable x_j .

$$\begin{aligned} \partial_j |x|^a &= \frac{a}{2} (x_1^2 + \dots + x_n^2)^{\frac{a-2}{2}} \cdot 2x_j. \\ &= ax_j |x|^{a-2}. \end{aligned}$$

Which is of the desired form.

Now, suppose some partial of $|x|^a$ is of the form $\partial^\alpha |x|^a = \sum_{i \in I} C_i x^{\alpha_i} |x|^{P_i}$. Then

$$\partial_j \sum_{i \in I} C_i x^{\alpha_i} |x|^{P_i} = \sum_{i \in I} \partial_j C_i x^{\alpha_i} |x|^{P_i}.$$

So we must only show that $\partial_j C_i x^{\alpha_i} |x|^{P_i} = C_i^* x^{\alpha_i^*} |x|^{P_i^*}$. We proceed by cases.

Case 1: Suppose x_j appears in x^{α_i} with power k . Define $\alpha_i^* = \alpha_i - (0, \dots, k, \dots, 0)$, where k appears in the j th entry. Then

$$\begin{aligned} \partial_j C_i x^{\alpha_i} |x|^{P_i} &= \partial_j C_i x^{\alpha_i^*} x_j^k |x|^{P_i}. \\ &= C_i x^{\alpha_i^*} [k x_j^{k-1} |x|^{P_i} + \frac{P_i}{2} (x_1^2 + \dots + x_n^2)^{\frac{P_i-2}{2}} \cdot 2x_j^{k+1}]. \\ &= C_i x^{\alpha_i^*} [k x_j^{k-1} |x|^{P_i} + P_i x_j^{k+1} |x|^{P_i-2}]. \end{aligned}$$

Let $\beta = \alpha_i - (0, \dots, 1, \dots, 0)$, where 1 is in the j th entry, $\gamma = \alpha_i + (0, \dots, 1, \dots, 0)$ with 1 in the j th entry.

$$= (C_i k) x^\beta |x|^{P_i} + (C_i P_i) x^\gamma |x|^{P_i-2}. \quad (1)$$

Which is of the desired form. If we take the case where x_j does not appear in the multi-index, the first term of (1) vanishes, which still results in the desired form. In each case, each part of the sum has order reduced by 1. $\square$

Notice that the form shown in Lemma 1 is continuous for all $x \neq 0$. Therefore, behaviour at 0 is the defining factor of whether or not $|x|^a \in \mathcal{C}^k(\mathbb{R}^n)$. We now prove a second lemma showing that if $\text{ord}(C_i x^{\alpha_i} |x|^{P_i}) > 0$, then $C_i x^{\alpha_i} |x|^{P_i}$ is continuous at $x = 0$.

Lemma 2. *If $\text{ord}(C_i x^{\alpha_i} |x|^{P_i}) > 0$, then $C_i x^{\alpha_i} |x|^{P_i}$ is continuous at $x = 0$.*

Proof. Suppose $\text{ord}(C_i x^{\alpha_i} |x|^{P_i}) > 0$. If both $|\alpha_i|, P_i \geq 0$, then clearly $C_i x^{\alpha_i} |x|^{P_i}$ is continuous at $x = 0$. Notice that $|\alpha_i|$ cannot be less than 0 by definition, so suppose that $|P_i| < 0$. Then $|\alpha_i| > |P_i|$. Then

$$\begin{aligned} |C_i x^{\alpha_i} |x|^{P_i}| &= |C_i| \frac{|x^{\alpha_i}|}{|x|^{|P_i|}} \\ &\leq |C_i| \frac{|x|^{|\alpha_i|}}{|x|^{|P_i|}} \\ &= |C_i| |x|^{|\alpha_i| - |P_i|}. \end{aligned}$$

where $|\alpha_i| - |P_i| > 0$. So, $\lim_{x \rightarrow 0} |C_i x^{\alpha_i} |x|^{P_i}| = 0$ by the squeeze theorem. So $C_i x^{\alpha_i} |x|^{P_i}$ is continuous at $x = 0$. The inequality comes from equation 1.7.1 in the text. $\square$

As shown in Lemma 1, taking k partials will drop the order by k . So, for $k \leq [a]$, $\text{ord}(\partial_j |x|^a) \geq a - [a] > 0$. Thus, $|x|^a$ is continuously differentiable $[a]$ times. We have shown that $|x|^a \in \mathcal{C}^{[a]}(\mathbb{R}^n)$. We can show that $|x|^a \notin \mathcal{C}^{[a]+1}(\mathbb{R}^n)$ by counterexample.

Let $x = (x_1, 0, \dots, 0)$. Then

$$\begin{aligned} \partial_1 |x|^a &= a |x_1|^{a-1} \text{sign}(x_1). \\ \partial_1^2 |x|^a &= a(a-1) |x_1|^{a-2} \text{sign}(x_1)^2. \\ &\vdots \\ \partial_1^k |x|^a &= a(a-1) \dots (a-(k-1)) |x_1|^{a-k} \text{sign}(x_1)^k. \end{aligned}$$

When $k = [a] + 1$.

$$\begin{aligned} \partial_1^{[a]+1} |x|^a &= a(a-1) \dots (a-[a]) |x_1|^{a-[a]-1} \text{sign}(x_1)^{[a]+1}. \\ &= a(a-1) \dots (a-[a]) \cdot \text{sign}(x_1)^{[a]+1} \frac{1}{x_1^{1+[a]-a}}. \end{aligned}$$

Which does not converge as $|x| \rightarrow 0$, so $\partial_1^{[a]+1} |x|^a$ is not continuous. Therefore $|x|^a \notin \mathcal{C}^{[a]+1}(\mathbb{R}^n)$. $\square$